

Assignment Overview

Project Assignment

In this unit, a project assignment is one of the main assessments. It is about the design and implementation of a simple intelligent system to solve a practical problem. Object-oriented programming at an intermediate level is the assumed knowledge to carry out this project.

There are three project titles to choose from:

- [Automated Negotiation System](#) [Automated Negotiation System Project Details.pdf]
 - **Project:** Design and implementation of a simple *multi-agent automated negotiation platform* involving agents representing car dealers, car buying agents and a broker agent.
 - **Modules involved:** Interaction Protocols, Automated Negotiation, Concurrent/multi-attribute negotiation.
- [Delivery Vehicle Routing System](#) [Delivery Vehicle Routing System Project Details.pdf]
 - **Project:** Design and implementation of a simple *delivery vehicle routing system* involving delivery agents and a master routing agent whose job is to find the optimal routes for the delivery vehicle agents to get their parcels to the receivers.
 - **Modules involved:** Interaction Protocols, Search/Optimization, Dynamic Adaptation
- [Automated Fall Detection using AI Vision](#) [Automated Fall Detection using AI Vision Project Details.pdf]
 - **Project:** This project develops an AI vision-based system to automatically detect human falls from video data using machine learning and deep learning techniques. Students will design computer vision models to distinguish fall events from normal daily activities under real-world conditions.
 - **Modules involved:** Machine learning, Deep Learning, Computer Vision

Due Date: Week 12 (Friday, 11:59PM)

Weighting: 50%

Group member and project registration: [here](#)

Project Requirements

- Source code maintained on Git-based VCS (Github/Bitbucket/GitLab/...). You must provide read-only access to the tutor/lecturer.
- Running illustrative demo of a working prototype (please refer to **Marking Scheme** for details on functionality that needs to be implemented).

- Project report (8-10 pages). Download the 'Who did what' declaration form [here](#) [COS30018 - Assignment Deliverable Guide.docx]
- Presentation + demo (pre-recorded video).